Common Rehabilitation Conditions HAS4185
復康科常見疾病 HAS4185J

L1: Orthopaedic – Joint Problems: Osteoarthritis, Rheumatology, Ankylosing Spondylitis, Joint Replacement

L1: 常見骨科疾病 – 關節問題: 骨關節炎, 類風濕性疾病, 強直性脊椎炎, 關節置換術

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Arthritis Overview
關節炎概述

Arthritis 關節炎

- **Latin word:**
  - arthron – Joint
  - itis – Inflammation
- **very common but NOT well understood**
- **NOT a single disease**
- **informal way of referring to joint pain / joint disease**
- **> 100 different types of arthritis & related conditions**

- **拉丁詞:**
  - arthron – 關節
  - itis – 發炎
- 常見但不清楚成因
- 並非單一疾病
- 關節痛或關節問題的非正式用語
- **> 100** 種不同類型的關節炎和相關疾病

Common signs & symptoms
常見體徵及症狀

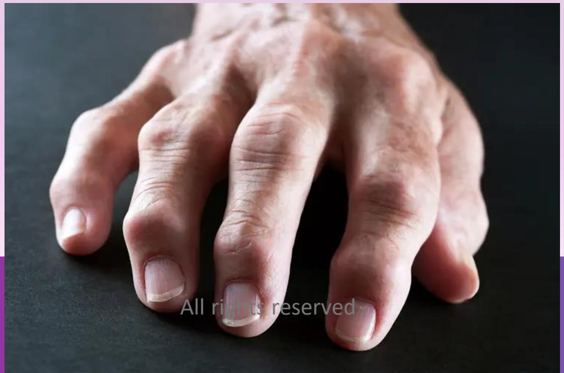
- **Joint swelling, pain, stiffness & decreased range of motion (ROM)**
- may last for years, progress / get worse with time
- mild, moderate or severe
- may result in chronic pain, inability to do daily activities, difficulty in walking (both level ground / stairs)

- **關節腫脹、疼痛、僵硬和活動範圍 (ROM) 減少**
- 可能持續數年，隨著時間的推移而惡化
- 輕度、中度或重度
- 可能導致慢性疼痛、無法進行日常活動、行走困難 (包括地面/樓梯)

Common signs & symptoms
常見體徵及症狀

- may cause permanent joint deformity (changes in shape)
- some are visible, such as knobby finger joints, but often only be seen on X-ray
- some types of arthritis also affect other areas like heart, eyes, lungs, kidneys & skin

- 可能導致永久性關節畸形 (形狀變化)
- 有些是可見的，例如腫大的手指關節，但通常只能在 X光上看到
- 某些類型的關節炎還會影響其他部位，如心臟、眼睛、肺、腎臟和皮膚



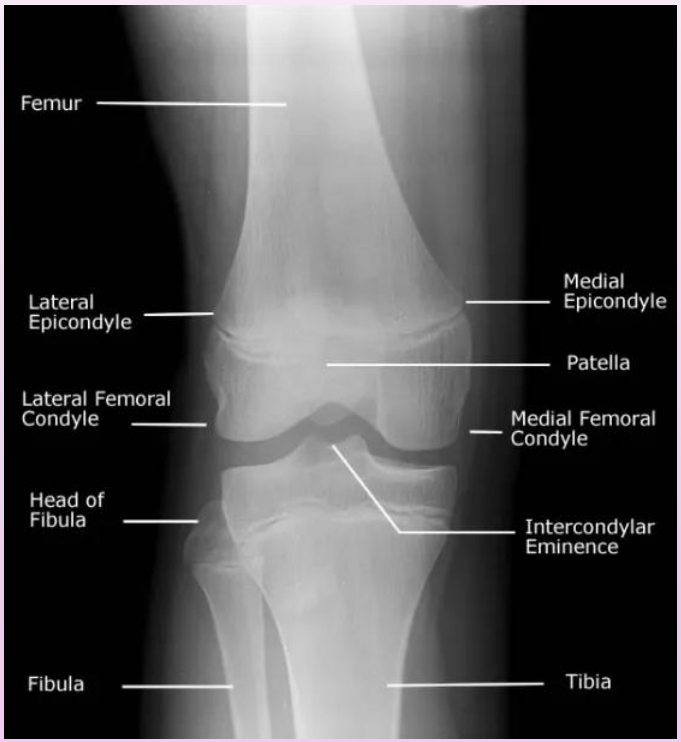
Diagnosis 診斷

- Subjective examination - onset, pattern, means to relieve
- Objective examination:
  - Acute inflammation:
    - **Redness, joint swelling, warmth & pain**
  - Joint crepitus: clicking / popping
  - "Locking" / "sticking" during movement
  - Range of motion (active, assisted, passive)
  - Joint instability with muscle weakness / buckling
- Imaging test: X-ray

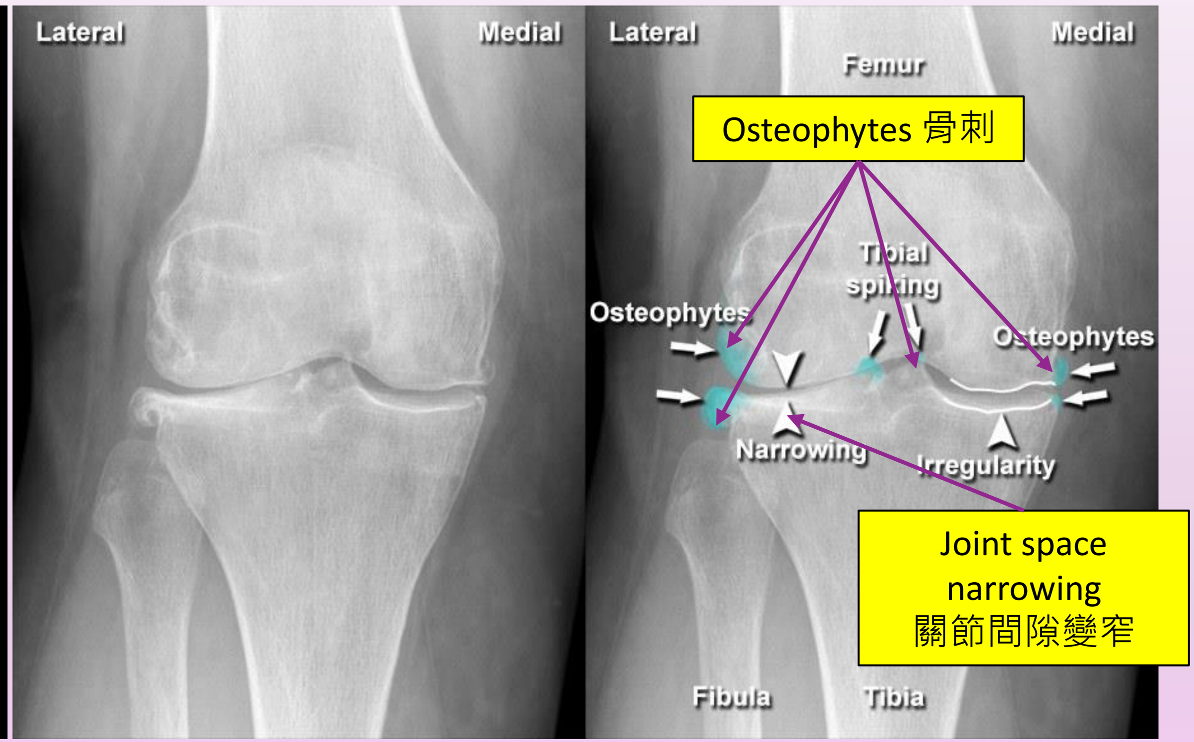
- 問診 – 發病時間、症狀表現模式、減輕症狀的方法
- 客觀檢查:
  - 急性炎症
    - **急性發炎 發紅、關節腫脹、發熱和疼痛**
  - 關節磨擦音: 咔噠聲/噼啪聲
  - 關節活動時 "鎖" / "粘"
  - 關節運動範圍 (主動, 輔助, 被動)
  - 關節不穩定伴肌肉無力 / 屈曲
- 影像學檢查: X光

X-ray X光

narrowing of joint space 關節間隙變窄
formation of bone spurs / osteophytes 骨刺的形成



X-ray of normal knee joint
正常膝關節的X光



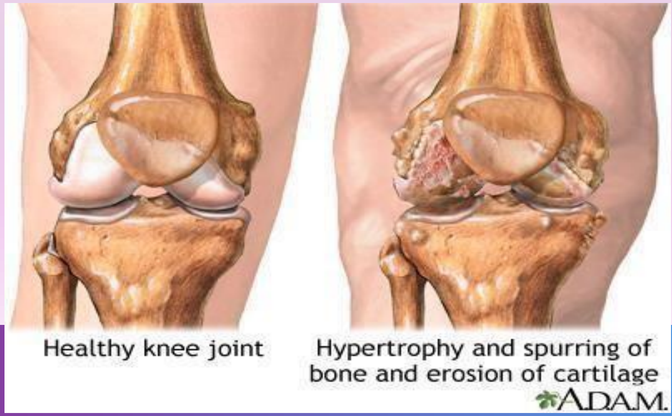
X-ray of OA knee joint
膝關節退化性關節炎的X光

Osteoarthritis
退化性關節炎

Osteoarthritis 退化性關節炎

- **Degenerative joint disease / "wear & tear" arthritis**
- Common chronic joint conditions
- Aged ≥50
- Commonly affected sites:
  - Weight bearing joints, e.g. knees, hips, low back & neck
  - Small joints of fingers, base of thumb & big toe

- **退行性關節病 / "磨損" 關節炎**
- 常見的慢性關節疾病
- 年齡 ≥50歲
- 常見受影響部位:
  - 負重關節, 例如膝, 髖, 下背部和頸部
  - 手指小關節, 拇指根部和大腳趾



Global and Local Prevalence
全球和本地患病率

- **Global: 595 million had OA in 2020 (i.e. 7.6%)**

  [GBD 2021 Osteoarthritis Collaborators, 2023]

- **Hong Kong:**
  - OA Knee:
    - Male: 7%
    - Female: 13%
  - OA Hip:
    - < 1% in older adults

  [Lau EC, 2000]

- **全球: 2020 年有 5.95 億人患有 OA (即 7.6%)**

  [GBD 2021 Osteoarthritis Collaborators, 2023]

- **香港:**
  - 膝關節退化性關節炎:
    - 男性: 7%
    - 女性: 13%
  - 髖關節退化性關節炎:
    - 老年人< 1%

  [Lau EC, 2000]

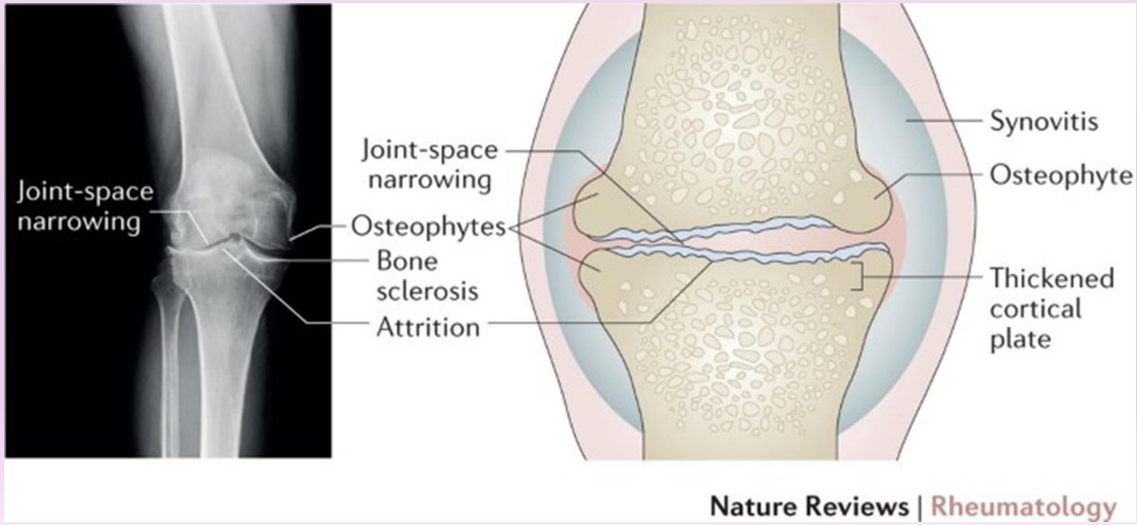
Progression of Osteoarthritis
退化性關節炎的發病機制

Break down of cartilage, affecting bone & synovium
軟骨分解，影響骨骼和滑膜

Decrease in protection between bones
骨骼之間的保護減少

Bone rubbing on bone
骨骼互相摩擦

Pain, swelling & problem with joint movements
關節運動時有疼痛、腫脹和活動問題

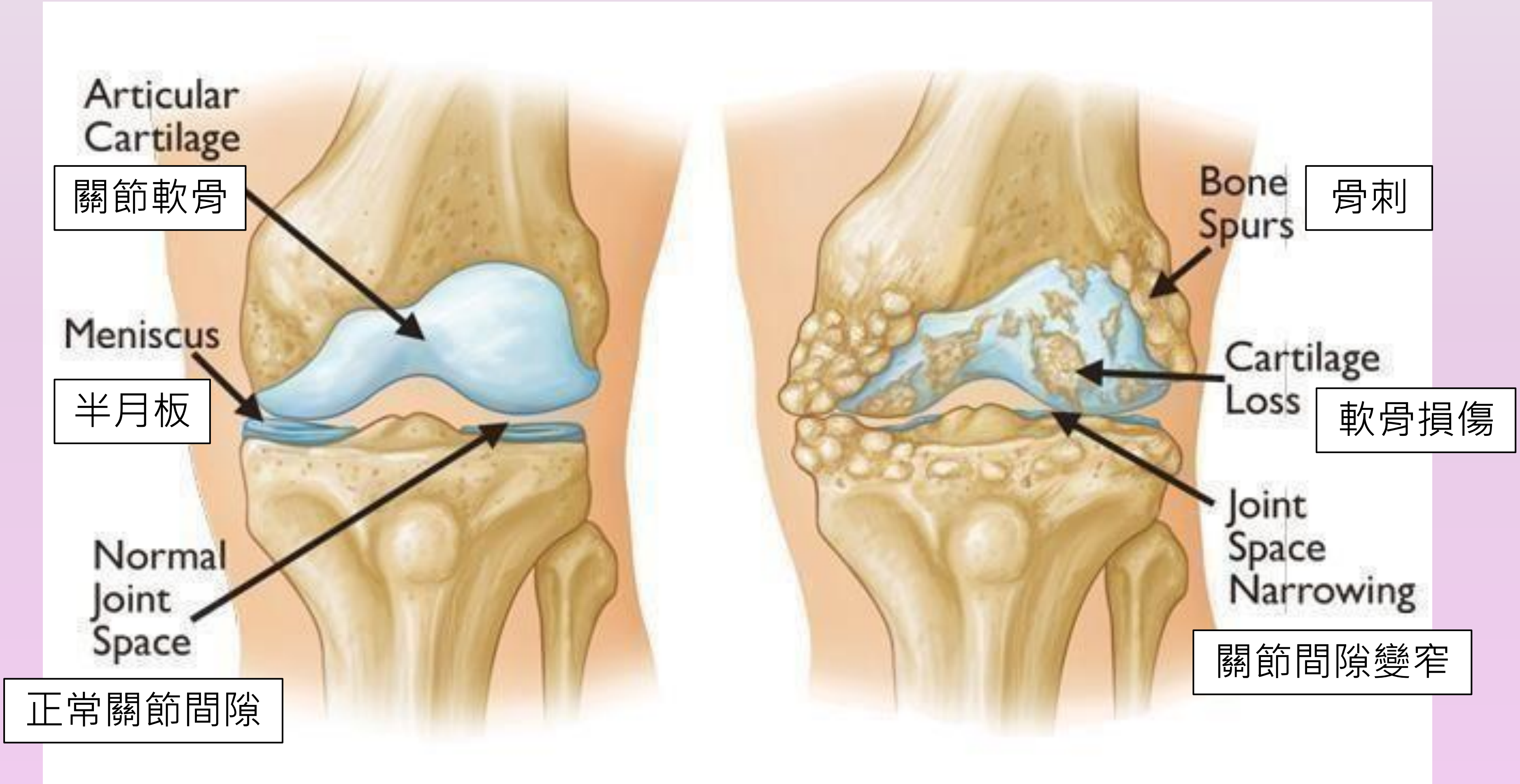


Osteoarthritis Knee (OA Knee)
膝退化性關節炎 (OA Knee)

- **Osmosis: Osteoarthritis - causes, symptoms, diagnosis, treatment & pathology:**
  https://youtu.be/sUOlmI-naFs?si=HTeZLYKHnbiQfEH9

- **杏林在線 (退化性關節炎):**
  https://www.youtube.com/watch?v=Vejj-XK7f6s

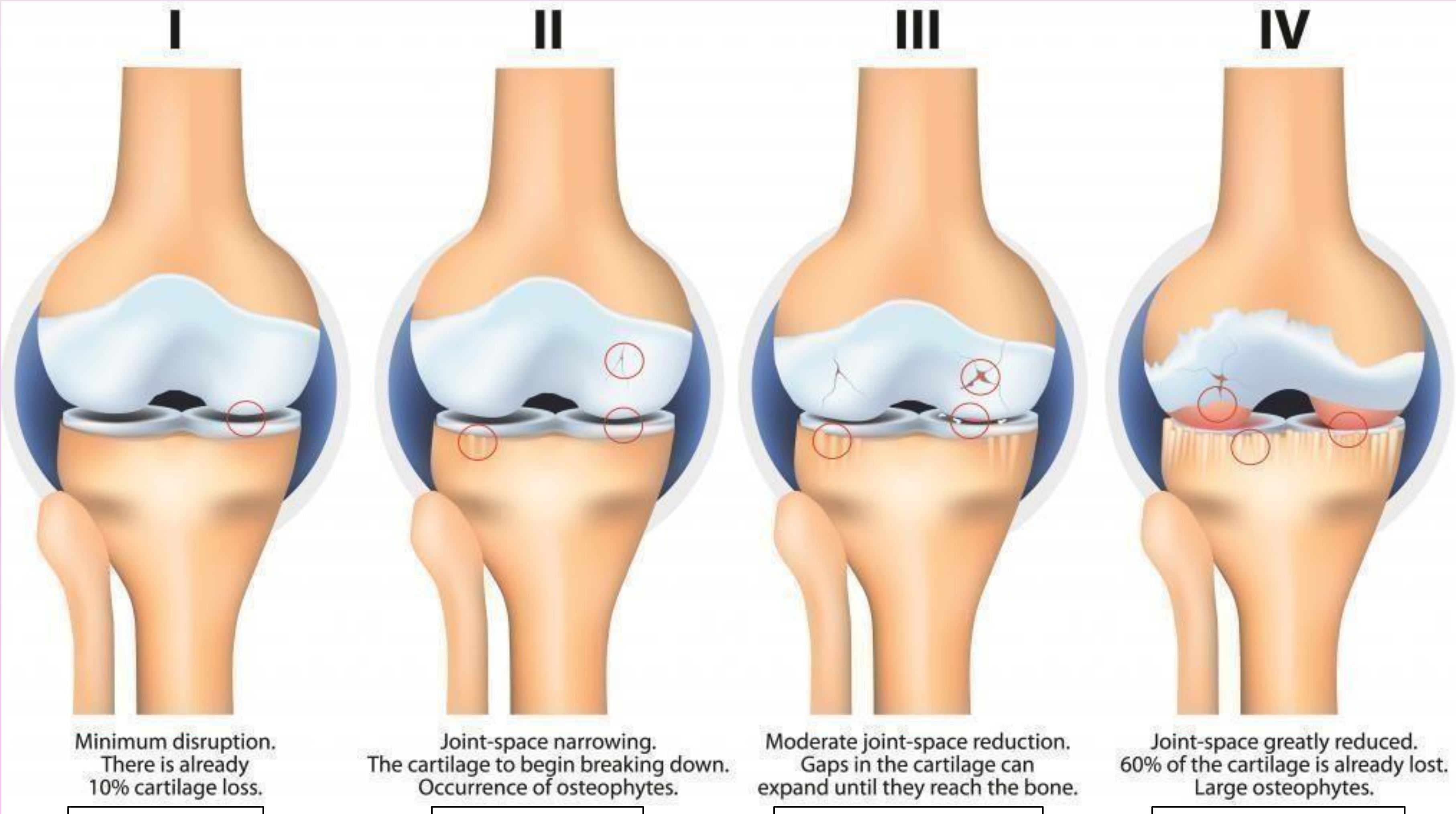
Normal Knee Joint 正常膝關節	OA Knee 膝關節退化性關節炎



Risk factors
風險因素

- Increasing age / aging
- Obesity
- Previous joint injuries
- Overuse
- Weak muscles
- Genetic factors

- 年齡增加 / 衰老
- 肥胖
- 曾有關節損傷
- 過度勞損
- 肌肉無力
- 遺傳因素



破壞最小，已經有 10% 的軟骨損失	關節間隙變窄，軟骨開始分解，骨刺增生	關節間隙已有中度減小，軟骨的縫隙擴大，以致骨骼露出	關節間隙大大減小，60%的軟骨已經損失，骨刺變大

Stage of Osteoarthritis Knee
膝退化性關節炎的階段

Stage 1: Minor

- slight damage to cartilage +/- osteophytes
- NO apparent narrowing of joint space

  (normal at X-ray)
- Asymptomatic (NO pain / discomfort)

第1階段: 輕微

- 軟骨輕微損傷 +/- 骨刺
- 關節間隙無明顯變窄 (X光正常)
- 無癥狀 (無疼痛 / 不適)

Stage 2: Mild

- some symptoms noted like joint stiffness & pain, esp. after prolonged sitting
- X-ray: clearly shown growth of osteophytes, cartilage begin to get thin

第2階段: 輕度

- 開始有癥狀，如關節僵硬和疼痛，尤其是在長時間坐著後
- X光: 清晰顯示骨刺生長，軟骨開始變薄

Stage of Osteoarthritis Knee
膝退化性關節炎的階段

Stage 3: Moderate

- moderate narrow joint space, with cartilage loss
- thickening of bone just underneath cartilage
- pain & discomfort with daily activities like walking, bending
- joint inflammation - excess synovial fluid
- increase swelling

第3階段：中度

- 關節間隙發展至中度狹窄，並有軟骨損失
- 軟骨下的骨骼增厚
- 走路、彎曲等日常活動時感到疼痛和不適
- 關節炎症 - 滑液過多
- 增加腫脹

Stage of Osteoarthritis Knee
膝退化性關節炎的階段

Stage 4: Severe

- further joint space narrowing, with continuous cartilage loss
- more significant pain & discomfort with daily activities due to joint friction
- X-ray: bone on bone (cartilage completely worn / little left)
- Deformity with varus (bow leg) or valgus (knock knee)

第4階段：嚴重

- 關節間隙進一步變窄，軟骨持續損失
- 由於關節摩擦導致的日常活動的疼痛和不適更嚴重
- X光: 骨骼對骨骼 (軟骨完全磨損 / 所剩無幾)
- 膝部畸形: 膝內翻 (O形腿) 或膝外翻 (X形腿)

Knee Deformity
膝關節畸形

Genu Varus
膝內翻

Genu valgus
膝外翻

Principles of Rehabilitation
復康原則

- Manage symptoms such as pain, stiffness & swelling
- Improve joint mobility & soft tissues flexibility
- Improve muscle strength
- Maintain healthy weight
- Sufficient exercise: recommended 150 minutes moderate intensity exercise weekly
- Strategies to reduce wear & tear of joints
- Use of assistive device, e.g. walking aids, knee brace, shoe orthotics, etc. to facilitate functions & mobility

- 控制疼痛、僵硬和腫脹等癥狀
- 改善關節活動度和軟組織柔軟度
- 提高肌肉力量
- 保持健康的體重
- 充足的運動：建議每周進行150分鐘的中等強度運動
- 減少關節磨損的策略
- 使用輔助設備，如助行器、護膝、鞋子矯形墊等，促進功能和活動的工具

Medications
藥物治療

Analgesics

- Non-opioid: e.g. paracetamol
- Non-steroidal anti-inflammatory drugs (NSAIDs) for inflammation & pain relief: e.g. aspirin, ibuprofen, naproxen
- Opioid analgesic: e.g. tramadol

Corticosteroids – oral / injection

- powerful anti-inflammatory

Hyaluronic acids - injection

- normally found in joint fluid, as shock absorber & lubricant decrease in OA joint

鎮痛藥

- 非鴉片類止痛藥，如撲熱息痛
- 非類固醇消炎藥 (NSAID)，用於緩解炎症和疼痛，例如阿士匹靈、布洛芬、萘普生
- 鴉片類止痛藥，如曲馬多

皮質類固醇 - 口服或注射

- 強效消炎藥

透明質酸 - 注射

- 退化關節一般會減少作為吸震及潤滑劑的關節液，注射透明質酸作為替代

Disease Management
疾病管理

Stage 1

- Exercise for muscle strengthening & joint mobility
- Encourage sufficient physical activity
- +/- analgesic medication (pills / ointment)
- Taking supplement: e.g. glucosamine (effects on pain relieve & slow down degeneration not confirmed)

第一階段

- 肌肉強化和關節活動能力的運動鍛煉
- 鼓勵足夠的體能活動
- +/- 鎮痛藥 (藥丸 / 軟膏)
- 服用補充劑：葡萄糖胺 (或有止痛及減慢退化，但研究未能確定效果)

Disease Management
疾病管理

Stage 2

- +/- analgesic medication (pills / ointment)
- Physiotherapy:
  - Ex for muscles strengthening & flexibility, joint mobility
  - Hydrotherapy
- knee brace / shoe insole - reduce stress on joints

第二階段

- +/- 鎮痛藥（藥丸/軟膏）
- 物理治療:
  - 運動治療訓練肌肉力量、柔韌性及關節活動能力
  - 水療
- 護膝 / 鞋墊 - 減輕關節壓力

Disease Management
疾病管理

Stage 3

- Analgesic medication
- Physiotherapy:
  - Ex for muscles strengthening & joint mobility
  - Electrophysical modality therapy, e.g. TENS, IFT
  - Hydrotherapy
- Knee brace / shoe insole - reduce stress on joints
- Taking supplement: Glucosamine, Chondroitin

第三階段

- 鎮痛藥
- 物理治療:
  - 運動治療訓練肌肉力量、柔韌性及關節活動能力
  - 物理儀器治療，例如TENS、干擾波治療
  - 水療
- 護膝 / 鞋墊 - 減輕關節壓力
- 服用補充劑: 葡萄糖胺軟骨素

Disease Management
疾病管理

Stage 4

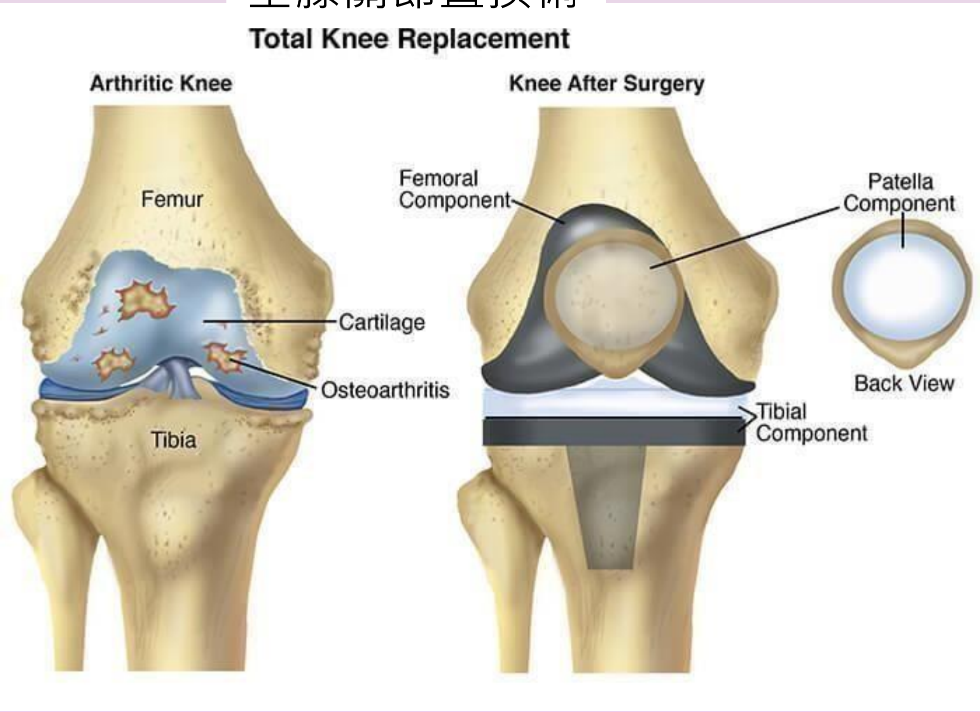
- Analgesic medication (pills / ointment / injection)
- Physiotherapy
- Knee brace / shoe insole - reduce stress on joints
- +/- walking aids
- Taking supplement: Glucosamine, Chondroitin
- Consider surgical intervention if failed conservative management

第 4 階段

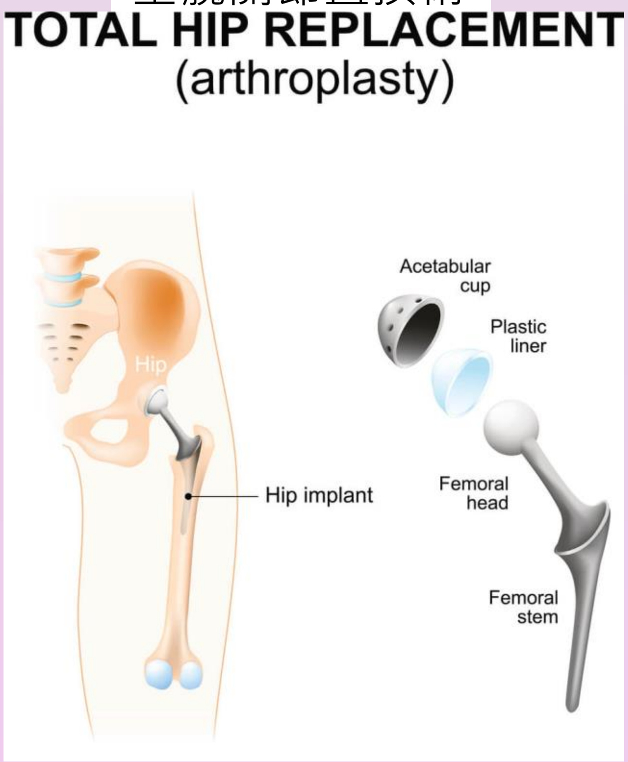
- 鎮痛藥 (藥丸/軟膏/注射劑)
- 物理治療
- 護膝 / 鞋墊 - 減輕關節壓力
- +/- 助行器
- 服用補充劑：葡萄糖胺、軟骨素
- 如果保守治療失效，考慮手術治療

Surgical Intervention
Total Joint Replacement
手術干預 - 全關節置換術

全膝關節置換術



全髖關節置換術



Prevention
預防

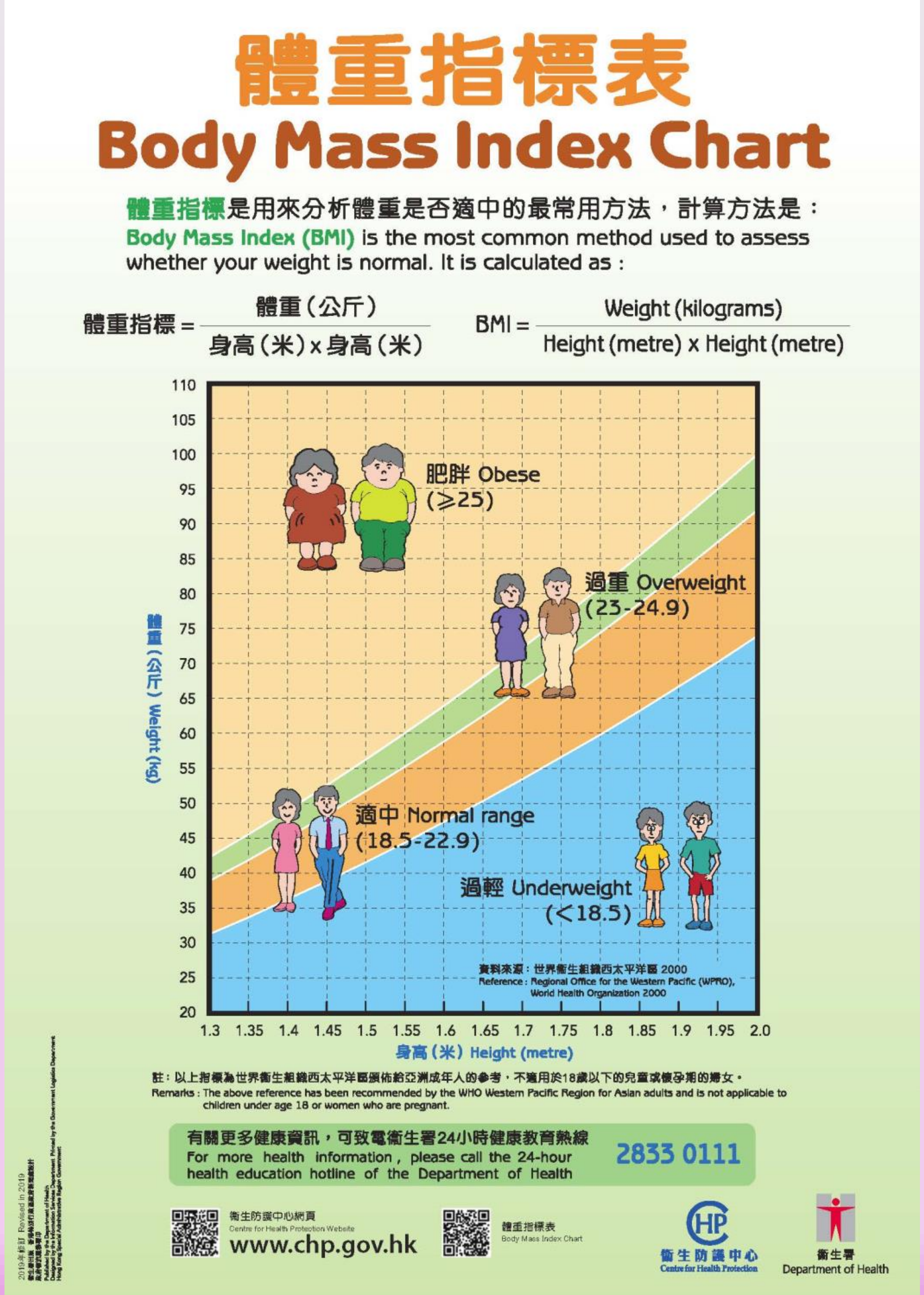
- **Weight management:**
  - Body Mass Index BMI (kg/m^2) – Asian
    - Underweight: <18.5
    - Normal: 18.5 to 22.9
    - Overweight: 23 to 24.9
    - Obesity: > 25

  (recommended by WHO Western Pacific Region for Asian adults)
- **Control blood sugar**
  - DM increases risk of inflammation
- **Regular exercise**
- **Minimize risk of injury & avoid overuse**

- **體重管理**
  - 體重指標 BMI (kg/m^2) – 亞洲人
    - 體重不足：<18.5
    - 正常：18.5 至 22.9
    - 超重：23 至 24.9
    - 肥胖：> 25

  (世界衛生組織西太平洋區域推薦給亞洲成年人)
- **控制血糖**
  - 糖尿病會增加炎症風險
- **恆常運動**
- **最大限度地降低受傷風險並避免勞損**

IVE 健康及生命科學 — HEALTH AND LIFE SCIENCES —



(Centre of Health Protection, 2019)

Rheumatoid Arthritis
類風濕性關節炎

Rheumatoid Arthritis (RA) 類風濕性關節炎

- **Chronic inflammatory autoimmune systemic disease**
  - Chronic – long term
  - Inflammatory – joint inflammation, thickening of synovium, leading to swelling, pain, +/- redness & warmth in joints
  - Autoimmune – immune system mistakenly attacks the joints
  - Systemic – affects entire body / multiple organ systems
- **Damage cartilage, bones & joints**
  - loss of cartilage, and narrow joint spacing loose, unstable joints with deformity

- **慢性炎症性自身免疫性系統性疾病**
  - 慢性 – 長期
  - 炎症 – 關節炎症，滑膜增厚，導致關節腫脹、疼痛、+/- 發紅和發熱
  - 自身免疫性 – 免疫系統錯誤地攻擊關節
  - 系統性 – 影響整個身體/多個器官系統
- **損傷軟骨、骨骼和關節**
  - 軟骨損失，關節間隙狹窄，關節鬆動及不穩定，關節畸形

Global and Local Prevalence
全球和本地患病率

- **Global:**
  - around 1% in the world

    [Radu & Bungau, 2021]
  - Up to 3% among older people

    [Turk, Liu & Pope, 2023]
- **Hong Kong:**
  - 0.35%
  - Women has 2-3 times higher prevalence than men

    [Lau E, 1993]

- **全球:**
  - 全球約1%

    [Radu & Bungau, 2021]
  - 老年人中高達 3%

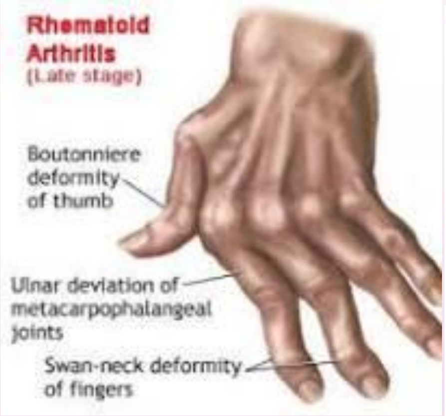
    [Turk, Liu & Pope, 2023]
- **香港:**
  - 0.35%
  - 女性的患病率是男性的 2-3 倍

    [Lau E, 1993]

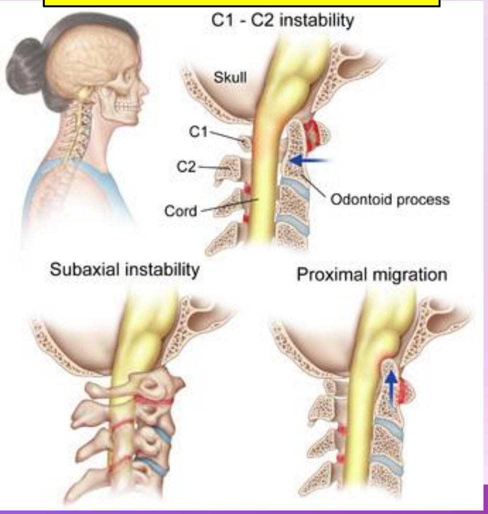
Common Joint Deformities in RA
類風濕性關節炎常見關節變形

Ulnar drift 尺側偏移

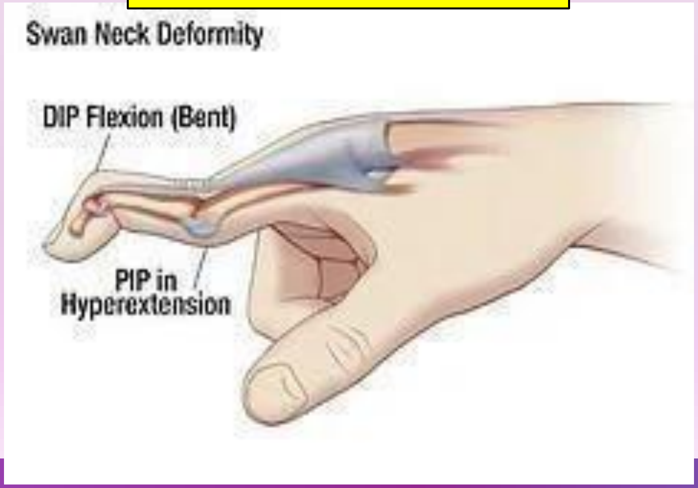
Claw toes 爪狀趾



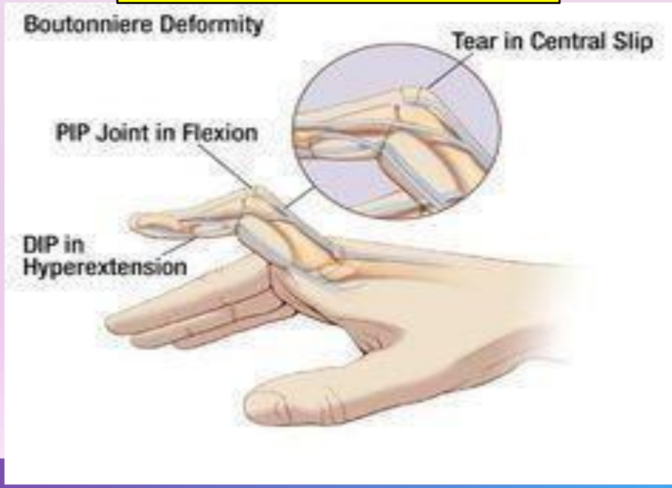
C1-C2 subluxation
第1、2頸椎半脫位



Swan Neck Deformity
天鵝頸變形



Boutonniere Deformity
鈕扣孔變形



Rheumatoid Arthritis (RA)
類風濕性關節炎

- Recommend early diagnosis with aggressive treatment
- most commonly affects joints of the hands, feet, wrists, elbows, knees and ankles
- usually symmetrical
- Can affect body systems, such as skin, eyes, cardiovascular or respiratory systems
- Can cause long-term disability, pain & premature death

- 建議通過早期診斷以及早積極治療
- 最常影響手、腳、手腕、肘部、膝蓋和腳踝的關節
- 通常影響對稱的關節
- 會影響身體系統，例如皮膚、眼睛、心血管或呼吸系統
- 可能導致長期殘疾、疼痛和過早死亡

Signs & symptoms – Joints
體徵及症狀 – 關節

- Joint pain, tenderness, swelling, redness or stiffness > 6 weeks
- Morning stiffness for > 1 hour
- > ONE joint are affected
- Small joints are commonly affected – Hands, Wrists & toes
- Symmetrical joint pain: on both sides of the body
- Decrease joint range of motion (ROM)
- Joint deformity

Flare - a period of high disease activity, last for days/months

**** increases in inflammation & other symptoms**

- 關節疼痛、壓痛、腫脹、發紅或僵硬 > 6 星期
- 晨僵 > 1小時
- 多於一個關節受到影響
- 小關節較常受影響 – 手指、手腕和腳趾
- 對稱性疼痛：身體兩側相同關節痛楚
- 關節變形
- 關節活動幅度減少

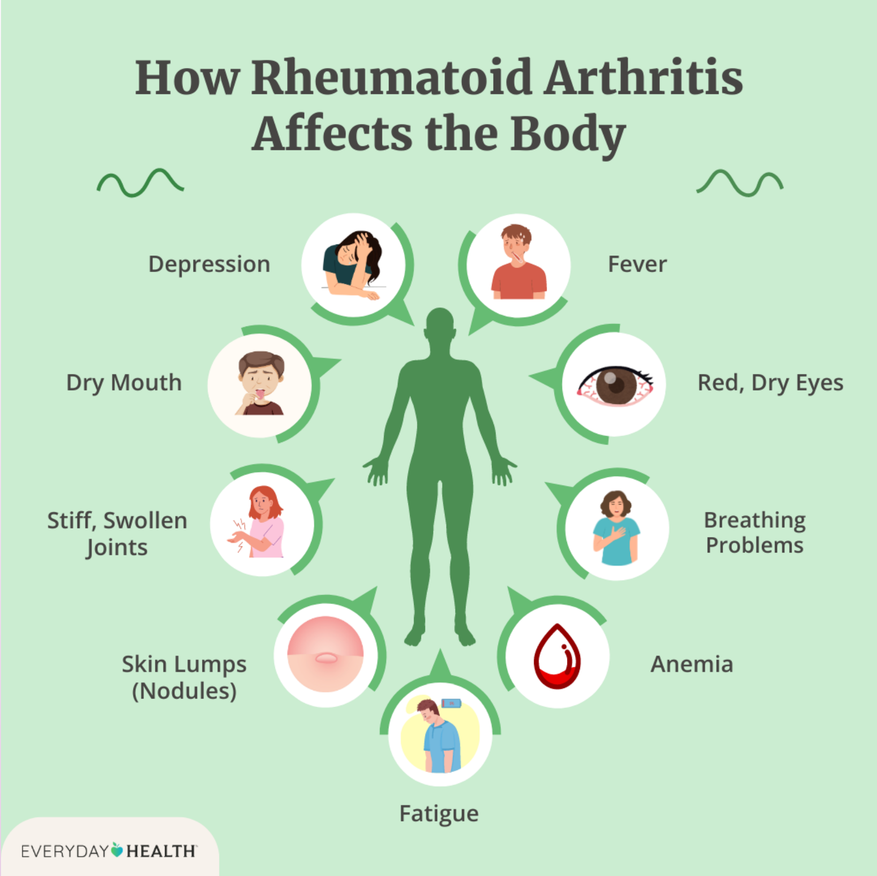
急性發作 - 疾病活動突然加劇，持續數天/數月

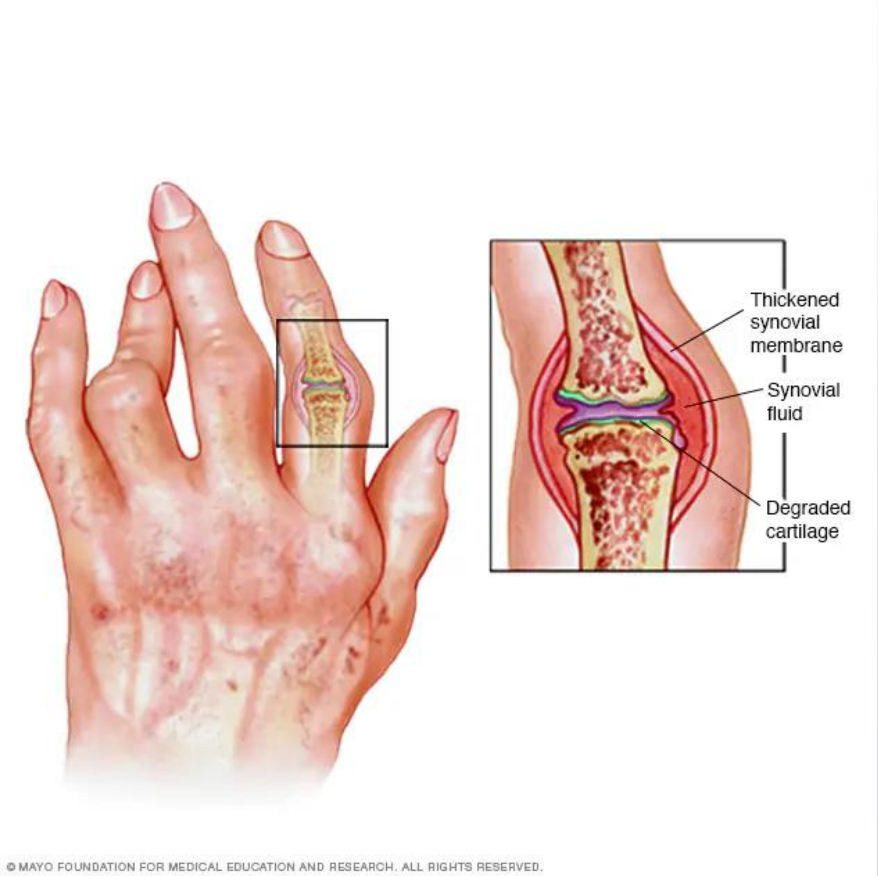
**** 炎症和其他癥狀增加**

Signs & symptoms – organs & body systems 體徵及症狀 – 其他器官及身體系統

- Whole body: fatigue, loss of appetite & low-grade fever
- Lungs: chest pain, shortness of breath (SOB), persistent cough - inflammation & scarring
- Heart: chest pain - inflammation of tissue around the heart
- Eyes: dryness, pain, redness
- Blood vessels: affect blood flow
- Skin: rheumatoid nodules – small lumps under the skin over bony areas
- Nerve: numbness, tingling, pain, weakness
- Blood: anaemia, low RBC
- Bone: Osteoporosis
- Mouth: dryness and gum problems

- 全身: 乏力、食欲不振、低燒
- 肺部: 胸痛、呼吸急促 (SOB)、持續咳嗽 – 肺部炎症和疤痕
- 心臟: 胸痛 - 心臟周邊組織發炎
- 眼睛: 乾燥、疼痛、發紅
- 血管: 影響血液流動
- 皮膚: 類風濕結節 – 近骨區域皮膚下的小腫塊
- 神經: 麻木、刺痛、疼痛、無力
- 血液: 貧血、低紅細胞
- 骨骼: 骨質疏鬆症
- 口腔: 乾燥和牙齦問題





Nucleus Medical Media – Rheumatoid Arthritis Animation

- https://www.youtube.com/watch?v=Yc-9dfem3lM

杏林在線 – 類風濕關節炎

- https://www.youtube.com/watch?v=dBnMDczmd7o

Risk factors
風險因素

- Increasing age
- Biological sex - female
- Genetic factors – HLA-DRB$_1$
- Infections (viral / bacterial)
- Gum disease
- Toxins
- Emotional stress
- Smoking
- Obesity

- 年齡增長
- 生理性別 - 女性
- 遺傳因素 – HLA-DRB$_1$
- 感染 (病毒 / 細菌)
- 牙齦疾病
- 毒素
- 情緒壓力
- 吸煙
- 肥胖

Diagnosis 診斷

Subjective examination:
- E.g. morning stiffness lasts for at least 1 hr, involvement of multiple joints, symmetrical

Blood test
- Check inflammation & biomarkers for RA
  - Inflammation markers
    - Erythrocyte Sedimentation Rate (ESR)
    - C-reactive Protein (CRP)
  - Antibodies
    - Rheumatoid factor (80% people with RA)

X-ray imaging:
- joint damage, bone erosion, narrow joint spacing

問診:
- 例如: 晨僵持續至少 1 小時, 涉及多個關節, 對稱性

血液檢驗:
- 檢查RA的炎症和生物標誌物
  - 炎症指標
    - 紅血球沉降率 (ESR)
    - 丙類反應蛋白檢驗 (CRP)
  - 抗體
    - 類風濕因子 (80% 的類風濕性關節炎患者會呈陽性)

X光影像：
- 關節損傷、骨骼侵蝕、關節間隙狹窄

Principles of Rehabilitation 復康原則

- Reduce / control inflammation
- Manage symptoms such as pain, stiffness & swelling
- Maintain joint ROM
- Joint protection strategies
- Sufficient exercise for maintenance of overall well-being: recommended 30-60 mins on 5 days / week with moderate intensity and exercise in water to reduce joint stress
- Use of assistive device, e.g. orthosis to decrease pain, prevent joint deformity
- Prevent long term complications
- Facilitate independency in ADLs

- 減少 / 控制炎症
- 控制症狀, 如疼痛、關節僵硬和腫脹等
- 維持關節活動幅度
- 關節保護策略
- 充足的運動以維持整體健康: 建議每周 5 天 30-60 分鐘，強度適中的運動，並在水中運動以減輕關節壓力
- 使用輔助設備，例如矯形器減輕疼痛及防止關節畸形
- 預防長期併發症
- 促進日常生活的獨立性

Examples of Orthosis & Assistive Devices
矯形器和輔助器具的例子

Hand Splint
手部矯形器

Anti-swan neck orthosis
防天鵝頸變形的矯形器

Door opening device
開門輔助工具

Foot orthosis
腳部矯形器

Medications
藥物治療

Analgesics

- Non-steroidal anti-inflammatory drugs (NSAIDs) for inflammation & pain relief: e.g. naproxen, diclofenac
- Cyclo-Oxygenase-2 (COX-2) inhibitors: e.g. celecoxib, etoricoxib

Disease-modifying anti-rheumatic drugs (DMARDs)

- Methotrexate, Leflunomide

Biologic DMARDs

- Anti-TNF α, B cell depletion therapy, anti-interleukin-6, CTLA-4, JAK inhibitors

Corticosteroids – oral / injection

- powerful anti-inflammatory

鎮痛藥

- 非類固醇消炎藥 (NSAID)，用於緩解炎症和疼痛, 例如萘普生、雙氯芬酸
- 環氧酶-2抑制劑 (COX-2), 例如塞來昔布、 依托考昔

改善病情抗風濕藥

- 甲氨蝶呤、來氟米特

生物製劑

- 抗腫瘤壞死因子、B細胞治療、抗第六介白素、T細胞協同刺激調節劑、JAK受體抑制劑

皮質類固醇 - 口服或注射

- 強效消炎藥

Disease Management
疾病管理

- Analgesic medication
- DMARDs
- Physiotherapy:
  - Ex for muscles strengthening & joint mobility
  - Electrophysical modality therapy, e.g. Wax therapy, TENS, IFT
  - Hydrotherapy
- Assistive device – prevent joint deformity
- Joint protection strategies, e.g. proper posture

- 鎮痛藥
- 改善病情抗風濕藥
- 物理治療:
  - 運動治療訓練肌肉力量、柔韌性及關節活動能力
  - 物理儀器治療, 例如蠟療、 TENS、干擾波治療
  - 水療
- 使用輔助器具 – 預防關節變形
- 關節保護措施, 如: 保持正確姿勢

Ankylosing Spondylitis
強直性脊椎炎

Ankylosing Spondylitis (AS)
強直性脊椎炎

- Relative uncommon rheumatic disease that affect the spine
  - Prevalence:
    - Global – 0.1-1.4%

      [Dean et al, 2014]
    - Hong Kong – 0.2-0.3%

      [Zhu et al, 2008]
  - long-term condition related to autoimmune disease
- Joints and ligaments of spine and other areas of the body become inflamed
  - e.g. sacroiliac joint, hips, shoulders
- May involve other organs such as eyes, lungs, heart

- 相對不常見影響脊椎的風濕性疾病
  - 患病率:
    - 全球 – 0.1-1.4%

      [Dean et al, 2014]
    - 香港 – 0.2-0.3%

      [Zhu et al, 2008]
  - 與自身免疫性疾病相關的長期病症
- 脊柱的關節和韌帶以及身體其他部位均發炎
  - 例如骶髂關節、臀部、肩部
- 可牽涉其他器官，如眼睛、肺、心臟

Signs & symptoms
體徵及症狀

- Symptoms tend to develop gradually (over several months or years)
- Morning stiffness > 30 min
- Symptoms may come and go over time (flare)
- Varies in symptoms, including pain, stiffness & swelling over back, sacroiliac joint, hips and shoulders
- extreme tiredness (fatigue)
- For severe cases, vertebrae fuse together lead to rigid spine

- 症狀往往逐漸發展 (數月或數年)
- 晨僵 > 30 分鐘
- 症狀可能會隨著時間的推移而出現和消失 (發作)
- 症狀各不相同, 包括和肩部疼痛、關背部、骶髂關節、臀部節僵硬和腫脹
- 極度疲倦 (疲勞)
- 嚴重的病例, 椎骨會融合在一起, 導致脊柱僵硬

Ankylosing Spondylitis (AS)
強直性脊椎炎

Causes:

- no known cause
- link with a particular gene variant known as **HLA-B27**

原因:

- 原因不明
- 與稱為 **HLA-B27** 的特定基因變異有關

Risk factors:

- Family history and genetics
- Age (usually symptoms **develop before age 45**)
- Biological gender: **male**
- Autoimmune disease: psoriasis, Crohn’s disease, ulcerative colitis

風險因素:

- 家族史和遺傳學
- 年齡 (通常癥狀**在 45 歲之前出現**)
- 生理性別: **男性**
- 自身免疫性疾病: 牛皮癬、克隆氏症、潰瘍性結腸炎

Ankylosing Spondylitis (AS)
強直性脊椎炎

Treatment:
- No cure for AS, treatment available to relieve the symptoms and delay progression
- Medications to reduce pain & inflammation:
  - NSAIDS
  - DMARDs, Biologics
  - Corticosteriods
- Exercises: reduce pain & stiffness
- Physiotherapy: improve posture, flexibility, joint mobility
- Surgery: for rare cases to repair significantly damaged joints or correct severe bends in spine

治療:
- 無法治癒，可緩解症狀並延緩其進展
- 藥物以減輕疼痛和炎症:
  - 非類固醇消炎藥
  - 改善病情抗風濕藥，生物製劑
  - 皮質類固醇
- 恆常運動: 減輕疼痛和僵硬
- 物理治療: 改善姿勢、柔韌性、關節活動度
- 手術：在極少數情況下修復明顯受損的關節或矯正脊柱的嚴重彎曲

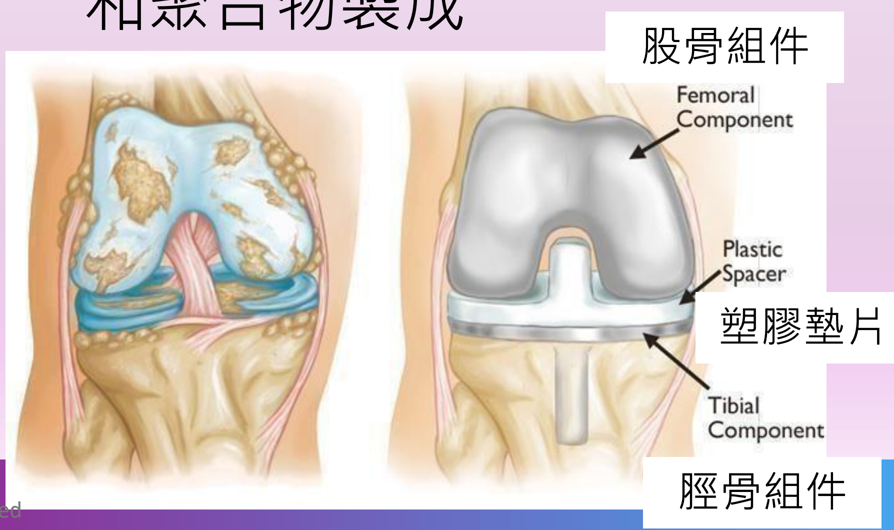
Total Joint Replacement
全關節置換手術

Total Knee Replacement (TKR)
全膝關節置換術

Total Knee Replacement (TKR)
全膝關節置換術

- Known as knee arthroplasty
- Mostly for OA knee that failed in conservative treatment (severe stage)
- Cut away damaged bone & cartilage from thigh, shin & knee cap, replace with an artificial joint
- made of metal alloys, high-grade plasters & polymers

- 稱為膝關節置換術
- 主要用於保守治療失敗的膝關節退化性關節炎(嚴重期)
- 切掉大腿骨、脛骨和膝蓋骨受損的骨骼和軟骨，以人工關節代替
- 由金屬合金、高級石膏和聚合物製成



Post-op Management
術後管理

Post surgery Day 1 - 7

- Pain & swelling control
- ROM ex
- Strengthening ex
- Weight bearing with assistance as tolerated
- Gradually increase WB to functional training
- Note for wound condition

Post surgery Week 1 - 3

- Continue phase 1 management as necessary
- ROM ex – goals for knee flex 95-115° & knee ext 0° in first few weeks
- Gradually increase walking and stair training

術後第 1 - 7 天

- 控制疼痛和腫脹
- ROM 運動訓練
- 肌肉強化運動
- 在可耐受的情形下, 由治療師輔助及指導負重訓練
- 逐漸增加負重, 以及功能訓練
- 注意傷口狀況

術後第 1 - 3 周

- 根據需要繼續第 1 階段
- ROM 運動訓練 – 目標於前幾週達到膝屈曲95-115°及膝伸直0 °
- 逐漸增加步行和樓梯訓練

Post-op Management
術後管理

Post surgery Week 3 – 6

- gradually resume most of daily activities, such as light housekeeping, shopping

術後第 3 – 6 周

- 逐漸恢復大部分日常活動，如輕度家務、購物

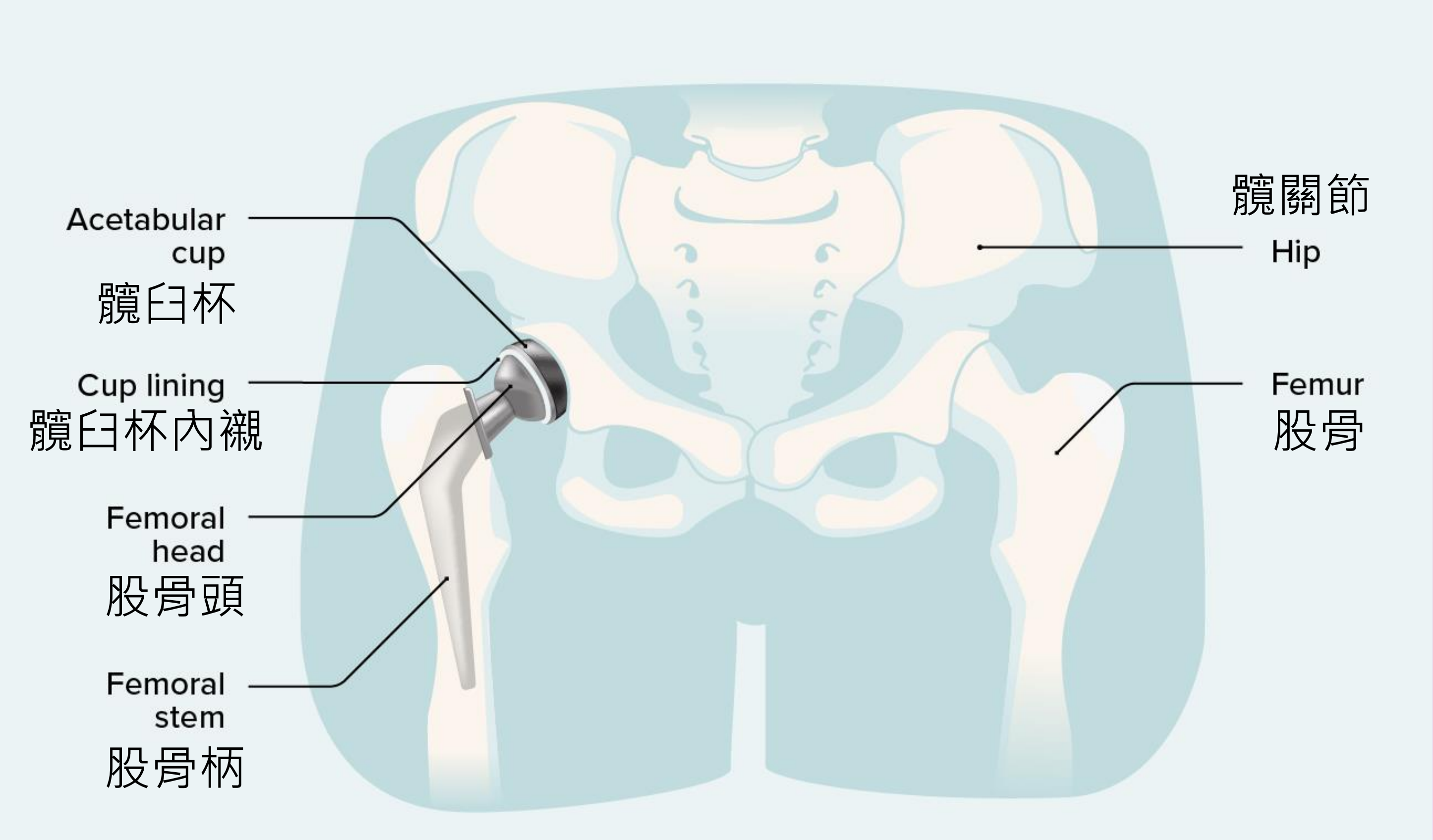
After recovery

- encourage more low impact activities like swimming, walking, cycling
- Avoid high impact ones like running, skiing, tennis or those with contact & jumping

恢復後

- 鼓勵更多低強度活動，如游泳、步行、騎自行車
- 避免跑步、滑雪、網球等高衝擊力活動，或碰撞和跳躍的活動

Total Hip Replacement (THR)
全髖關節置換術



Total Hip Replacement (THR) 全髖關節置換術

- Known as total hip arthroplasty
- In HK, 33-50.9% THR due to osteonecrosis (avascular necrosis)
- Cut away damaged femoral head (ball) and replace with a metal / ceramic component; also the acetabulum (socket) in the pelvis is resurfaced with a metal / plastic / ceramic liner

- 稱為髖關節置換術
- 在香港, 有33-50.9%的THR是由於骨枯 (缺血性壞死) 導致
- 切掉受損的股骨頭 (球) 並更換為金屬 / 陶瓷部件; 骨盆中的髖臼 (窩) 打磨後用金屬 / 塑膠 / 陶瓷作內襯

Post-op Management
術後管理

Post surgery Day 1 - 7

- Pain & swelling control
- Hip precautions: hip flex < 90°, no hip adduction, no hip rotations
- Positioning with Abductor pillow
- ROM ex
- Strengthening ex
- Weight bearing with assistance as tolerated
- Gradually increase WB to functional training
- Note for wound condition

術後第 1 - 7 天

- 控制疼痛和腫脹
- 髖關節注意事項: 髖關節屈曲 <90°, 不能髖關節內收, 不能髖關節旋轉
- 使用外展枕定位
- ROM 運動訓練
- 肌肉強化運動
- 在可耐受的情形下, 由治療師輔助及指導負重訓練
- 逐漸增加負重及功能訓練
- 注意傷口狀況

Post surgery Week 1 - 3

- Continue phase 1 management
- Gradually increase walking and stair training

術後第 1 - 3 周

- 繼續第一階段管理
- 逐漸增加步行和樓梯訓練

Post-op Management
術後管理

Post surgery Week 3 – 6

- gradually resume most of daily activities, such as light housekeeping, shopping

術後第 3 – 6 周

- 逐漸恢復大部分日常活動，如輕度家務、購物

After recovery

- encourage more low impact activities like swimming, walking
- Avoid high impact ones like running, skiing, tennis or those with contact & jumping
- Avoid cross legs in sitting
- Avoid sitting on low stool
- Recommend raised toilet seat
- NO static bike ex (as hip flex >90° during ex)

恢復後

- 鼓勵恢復低強度活動，如游泳、步行
- 避免跑步、滑雪、網球等高衝擊力活動，或碰撞和跳躍的活動
- 避免坐著時盤腿
- 避免坐在矮凳上
- 建議使用坐廁加高器
- 不可做腳踏單車運動 (因髖關節屈曲> 90°)

Precautions
注意事項

Abductor Pillow
外展枕

Avoid Cross Legs in Sitting
避免坐著時盤腿

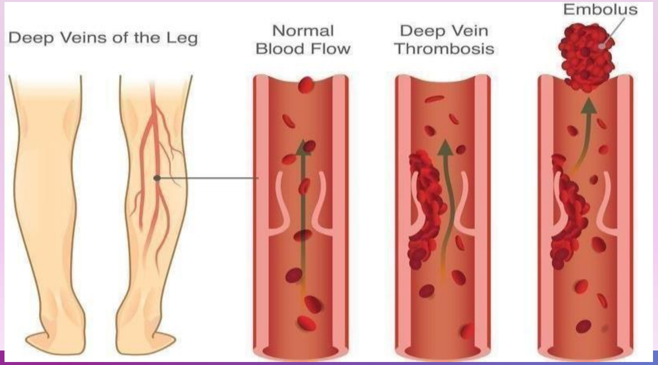
Raised Toilet Seat
坐廁加高器

Avoid Sitting on Low Stool
避免坐在矮凳

Risk of Surgery
手術風險

- Infection
- Deep vein thrombosis DVT (blood clots in vein)
- Nerve damage
- Heart attack / stroke
- Loosening of implants
- Bleeding
- Persistent pain and stiffness

- 感染
- 深層靜脈血栓 DVT (靜脈血栓)
- 神經損傷
- 心臟病發作 / 中風
- 植入物鬆動
- 出血
- 持續疼痛和僵硬



E-Resources

- Knee Osteoarthritis leaflet 膝關節退化性關節炎:
  - https://www3.ha.org.hk/ntwc/msc/docs/pep/22_23/%E7%89%A9%E8%86%9D%E8%89%AF%E6%A9%9F%20%E9%81%8B%E5%8B%95%E6%9C%89%E6%B3%95.pdf
- Exercise prescription recommendation for OA knee (English only) 膝關節退化性關節炎運動處方建議 (只有英語):
  - https://www.chp.gov.hk/archive/epp/files/DoctorsHanbook_ch9.pdf
- Total Joint Replacement Pre-operative information (全關節置換術術前資訊):
  - https://www3.ha.org.hk/ahnh/content/patientresources/physio/physio_eng/e_resource/TJR_Pre_eng/TJR_Pre_e.asp
- Total Joint Replacement Post-operative information (全關節置換術術後資訊):
  - https://www3.ha.org.hk/ahnh/content/patientresources/physio/physio_eng/e_resource/TJR_Mainpage_eng/TJR_Mainpage_eng.asp
- Hong Kong Arthritis & Rheumatism Foundation Ltd. 香港風濕病基金會:
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